Use Case Scenario 2. What do I need to do next

Who arrives. An adopter part way through. (This could be an individual, a project lead, a product manager, a technology team, an institutional decision-maker,a contributor working on a specific part of the adoption etc..) They have an AI-related project underway, believe they have made some progress, know roughly where they are and want to know what is coming.

What they want. What they now have to do, and for each of those, what others did and how it turned out. (eg. what they need to do next, if any prerequisites are missing, what they may have overlooked, what comparable adopters did, why they did it, what happened, whether the experience applies to their situation etc..)

Helped when. They see the next steps that arise at their point in the journey - including ones they had not realised they missed - and for each, what comparable adopters did, what happened, and what those adopters would tell them. A good understanding of their current position, the important unresolved issues, the most relevant next decisions/actions

Harmful when. The answer is generic. "Engage stakeholders early" is true, useless, and indistinguishable from advice they could have got anywhere. If what Cube returns does not reflect their situation, it adds nothing to what they already assumed.

1) Various Matching Scenarios

There could be many matching scenarios the CUBE might encounter and it needs to handle

	Match scenario	Possible Behaviour (To be discussed)
1	Exact pathway match	Use the direct experience. If Cube finds a previous project that is very similar to the adopter's situation, it should show what that adopter did, why they did it, what happened, and any conditions that mattered.

2	Partial match	Show what is similar and what is different. Cube should explain which parts of the previous experience are relevant and clearly point out the differences that may affect whether the learning can be applied.
3	Related use case	**Tell the user that this is a related, but not identical, experience.** Explain why it is relevant and what learning may still be useful, rather than presenting it as a direct precedent.
4	Evidence scattered across pathways	**Bring the relevant pieces together.** If useful information is spread across several pathways, Cube should combine the relevant evidence into one explanation while clearly showing which pathway or experience each point came from.
5	No evidence	**Be transparent about the gap.** If Cube cannot find relevant experience in the available knowledge, it should say so rather than inventing an answer or presenting generic advice as documented experience.
6	Conflicting evidence	**Show the different experiences and explain the conditions.** If different adopters took different approaches or got different results, Cube should show both and explain the circumstances in which each approach was used. It should not simply choose one.
7	Stale evidence	**Tell the user how old the evidence is.** Cube should show when the experience happened or was last verified and, where relevant, warn that technology, policies, operating conditions, or other factors may have changed.
8	User doesn't know stage	**Help the user figure out where they are.** Cube should ask a few simple questions about what they have already done, what they are currently testing or operating, and what remains unresolved, rather than requiring the user to understand the Explore/Define/Pilot/Scale terminology.
9	Missing critical context	**Ask only the important question that is needed.** If Cube cannot give a meaningful answer without

		knowing something important, it should ask for that information before giving guidance.
10	Missing non-critical context	**Don't unnecessarily stop the conversation.** If the missing information is unlikely to change the answer significantly, Cube can proceed using a reasonable assumption and clearly state the assumption it has made.
11	User has skipped earlier work	**Point out the possible missing prerequisite.** If the user's current position suggests that an important earlier decision or activity has not been addressed, Cube should flag it as something to check before moving forward. It should not simply say, “You skipped this.”
12	Multiple parallel workstreams	**Show the workstreams separately.** If, for example, the technology is in Pilot while institutional readiness is still in Define, Cube should represent these as separate areas rather than forcing everything into one overall stage.
13	User changes context	**Update the guidance based on the new information.** If the user provides new information that changes their situation, Cube should reconsider the relevant evidence and next steps rather than continuing with the previous assumptions.
14	Narrow technical question	**Understand enough of the bigger picture first.** If the user asks something like “Should we use RAG?”, Cube should first establish the minimum context needed to understand what problem they are solving and why, and then use relevant pathway evidence to answer.
15	Large number of relevant pathways	**Don't overwhelm the user with documents.** Cube should identify common patterns across the relevant pathways and present a concise synthesis, while allowing the user to drill down into the individual experiences and sources if needed.
16	No direct experience in the corpus but strong cross-domain pattern	**Explain that the evidence comes from a different domain.** If there is no directly comparable experience but the same type of

		problem or decision appears repeatedly in other domains, Cube can present that as related evidence and explain why it may be relevant.
17	Evidence has no documented outcome	**Don't imply that it worked.** Cube should say what the adopter did, if documented, but explicitly state that the source does not document the outcome. It should not infer success or failure from the absence of information.

2) Functional Requirements - what cube must be capable of doing

1	**Cube shall identify the context when the user's intent corresponds to Use Case 2** When the user asks something like “What should we do next?”, Cube should first understand the situation they are talking about - what they are building, who it is for, what has already happened, and where they currently are.
2	**Cube shall determine whether sufficient context exists before providing substantive guidance.** Cube should check whether it knows enough about the adopter's situation to give useful advice. If it does not, it should ask for the missing information instead of immediately giving generic recommendations
3	**Cube shall ask only questions that materially affect the guidance.** Cube should not conduct a long questionnaire. It should ask only questions where the answer could actually change what it recommends or shows next.
4	**Cube shall help / allow the adopter to correct or update previously established context.** The adopter should be able to say things like **“Actually, we haven't started the pilot yet”** or **“We changed our target users.”** Cube should update its understanding and adjust its guidance accordingly.
5	**Cube shall determine the adopter's position across Persona, Solution, Institution and Ecosystem.**

	Cube should understand where the adopter currently stands in each of the four dimensions - not just where they are technologically. For example, the solution may be in Pilot while institutional readiness is still in Define.
6	**Cube shall determine a stage for each dimension where sufficient evidence exists.** Where there is enough information, Cube should determine whether each dimension is in **Explore, Define, Pilot or Scale**. If there isn't enough evidence, it should leave the stage uncertain (or ask more questions to find that out) rather than guessing.
7	**Cube shall distinguish user-declared stage from evidence-derived stage.** Cube should distinguish between **"the adopter says they are in Scale"** and **"based on what they have told us, Cube believes they are still in Pilot."** These should not be silently treated as the same thing.
8	**Cube shall identify ongoing activities.** Cube should understand what the adopter is currently working on - for example, running a pilot, collecting data, testing with users, setting up monitoring, or negotiating with a partner.
9	**Cube shall identify completed activities.** Cube should keep track of what has already been completed so that it doesn't repeatedly recommend work the adopter has already done.
10	**Cube shall identify unresolved decisions.** Cube should identify important decisions that have not yet been settled - for example, who owns the system, whether human review is required, what success means, or which users to target.
11	**Cube shall identify missing prerequisites.** Cube should identify things that need to be in place before the adopter can reasonably proceed. For example, a scale-up decision may require operational ownership, monitoring, support processes or funding.
12	**Cube shall identify dependencies** Cube should understand what the adopter is waiting for or depends on - such as data access, another team, an external partner, an API, infrastructure, regulatory approval, or procurement.
13	**Cube shall identify potential missed steps.** Cube should compare the adopter's current situation with relevant pathway experience and flag important things that may not yet have been addressed. It should present these as **"potentially missed" or "worth checking"**, rather than assuming the adopter has made a mistake.
14	**Cube shall retrieve pathway evidence using adopter context.**

	Cube should search the pathway knowledge using the adopter's actual situation - problem, users, sector, stage, dimension, constraints, etc. - rather than simply matching keywords from the question.
15	**Cube shall support exact matching.** Cube should be able to find previous experiences that closely resemble the adopter's situation and use those experiences as direct evidence.
16	**Cube shall support partial matching.** Cube should also be able to find experiences that are similar but not identical, and clearly explain the differences that may matter.
17	**Cube shall support related-pattern matching.** Cube should be able to recognise when a different use case contains a similar underlying problem, decision or pattern that may still be useful to the adopter.
18	**Cube shall support cross-pathway retrieval.** Relevant evidence may exist across several pathway documents. Cube should be able to find and bring together those pieces rather than limiting itself to one pathway.
19	**Cube shall identify when no sufficiently relevant evidence exists.** If Cube cannot find useful documented experience, it should say so clearly rather than filling the gap with invented or generic “pathway” advice.
21	**Cube shall assess evidence applicability against the adopter's context..** Finding something similar is not enough. Cube should consider whether that experience actually applies to this adopter's situation before using it to support guidance.
22	**Cube shall not present materially different experiences as exact matches.** Cube must not make a loosely similar project sound like an identical one. If important differences exist, they should be visible to the user.
23	**Cube shall consider pathway conditions when determining applicability** Cube should consider **“when did this approach work?”** and **“under what conditions?”** For example, an approach may have worked because the adopter had dedicated staff or reliable data. Those conditions matter when deciding whether the learning applies elsewhere.
24	**Cube shall identify significant contextual differences between the adopter and the source experience.** Cube should point out important differences - such as different users, geography, organisation size, technology, regulations, resources or operating model - when those differences could affect the applicability of the experience.

25	**Cube shall distinguish direct documented evidence from synthesis.** Cube should make it clear whether something comes **directly from one documented pathway** or whether Cube has combined learning from several pathways to identify a broader pattern.
26	**Cube shall distinguish pathway evidence from external evidence (if cube support external evidence)** If Cube uses information from outside the pathway corpus, the user should be able to tell that it is external information and not experience documented in the Cube pathways.
27	**Cube shall distinguish evidence from Cube-generated inference.** Cube may sometimes draw a reasonable conclusion from available information, but it should make clear that this is **Cube's interpretation**, not something an adopter actually documented.
28	**Cube shall indicate when evidence may be stale.** Cube should show when an experience happened or was last verified, particularly when technology, regulations, policies or operating conditions may have changed since then.
29	**Cube shall not fabricate undocumented facts, outcomes, conditions or contributor statements.** If the source doesn't say something, Cube must not make it up. For example, it must not claim that an adopter **"increased adoption by 30%"** unless that outcome is actually documented.
30	**Cube shall identify potential next actions.** Cube should identify practical things the adopter may need to **do next** - for example, conduct another round of user testing or establish an operational support process.
31	**Cube shall identify potential next decisions.** Cube should identify decisions that need to be **made next** - for example, deciding who owns the system or whether human review is required.
32	**Cube shall identify potential next validations.**

	Cube should identify things that need to be **tested or verified before proceeding** - for example, validating accuracy with real users or checking whether a proposed workflow actually works in practice.
33	**Cube shall identify dependencies that need resolution.** If progress depends on something outside the adopter's immediate control, Cube should identify it and make it visible as something that needs to be resolved.
34	Cube shall explain why a proposed next step is relevant. Cube should not simply produce a checklist. For every important next step, it should explain **why this matters for this particular adopter at this point in their journey**
35	**Cube shall present comparable adopter experience where available.** Where relevant experience exists, Cube should show **what another adopter actually did**, rather than giving only generic recommendations.
36	**Cube shall present documented outcomes where available.** Cube should tell the adopter what happened after the previous adopter took that approach - including positive, negative or mixed outcomes - when the source documents them.
37	**Cube shall explain relevant conditions and limitations.** Cube should explain **when the experience may apply, when it may not, and what limitations or risks were observed.** This helps the adopter decide how much confidence to place in the experience.

3) Example Scenarios and Cube Behaviour - examples of how those requirements should appear in practice.

1	**Insufficient context** **Given** an adopter asks "What should I do next?" **When** Cube lacks information that materially affects the answer **Then** Cube asks targeted questions before producing detailed guidance.
2	**Exact match** **Given** a highly similar documented pathway exists **When** the adopter's context is established **Then** Cube presents the pathway as direct comparable evidence and explains the similarity.
3	**Partial match**

	Given a pathway matches the problem and stage but differs materially in geography **Then** Cube identifies the difference rather than presenting the pathway as an exact match.
4	**Cross-pathway evidence** **Given** no single pathway contains sufficient evidence **When** multiple pathways contain complementary relevant units **Then** Cube may synthesize them and identify the contributing pathways and clearly mention the same
5	**No evidence** **Given** no sufficiently relevant documented evidence exists **Then** Cube explicitly states that the evidence is missing rather than fabricating guidance.
6	**Missed step** **Given** the adopter believes they are ready for the next stage **When** important prerequisites appear unresolved **Then** Cube surfaces those unresolved areas and explains why they may matter.
7	**Conflicting evidence** **Given** two pathways document different approaches **Then** Cube presents the approaches and relevant conditions rather than arbitrarily selecting one. **Generic answer prevention** **Given** a next-step recommendation is generated **Then** the response should identify the contextual evidence, reasoning or explicitly labelled inference supporting it.
8	**Evidence provenance** **Given** a material recommendation is presented **Then** the adopter should be able to identify where the relevant experience came from.

4) Open questions / Pending Decisions

1. How much context should Cube remember?
Session only, project-level, organisation-level, or persistent adopter profile?

2. What qualifies as sufficient evidence?
How much evidence is required before Cube can present a pattern?

3. How should evidence quality be represented?
Simple categories or a more formal evidence model?

4. When should external search be allowed?
Always, only when requested, or automatically when pathway evidence is insufficient?

5. Should Cube recommend actions or only present evidence?
The current direction suggests contextual guidance, but the boundary needs to be explicitly defined.

6. How should conflicting pathways be represented?
Side-by-side evidence, conditions, or a structured decision matrix?

7. How should the system handle outdated pathways?
Retire, downgrade, flag, or continue using age indicators?